

E19898055_2

The equation $x^4 + 6x + y^3 + y - 1 = 0$ has no integer solutions

Size $h = 39$, length $l = 11.58$

Theorem 1. *The equation*

$$x^4 + 6x + y^3 + y - 1 = 0 \tag{1}$$

has no integer solutions.

The proof uses three polynomial identities of the form $H_k T_k = U_k^2 + d_k V_k^2$ on the curve (1), with $(d_1, d_2, d_3) = (2, 1, 2)$, and the fact that an odd prime p with $(-d_k/p) = -1$ divides $U_k^2 + d_k V_k^2$ only to an even power. Apart from polynomial identities, which can be checked by expanding both sides, and finite checks of congruences, the proof uses only basic properties of the Jacobi symbol and the law of quadratic reciprocity.

1 Jacobi symbols and the norm forms $a^2 + db^2$

Throughout this section, d is a squarefree integer with $d \neq 0, -1$; we apply the results with $d \in \{1, 2\}$. For a prime p and a non-zero integer N , we write $v_p(N)$ for the exponent of p in the prime factorization of N . For an integer a and an odd prime p , the Legendre symbol (a/p) is 0 if $p \mid a$, 1 if a is a non-zero square modulo p , and -1 otherwise. For an odd positive integer n , the Jacobi symbol is $(a/n) = \prod_p (a/p)^{v_p(n)}$, the product over the prime factors p of n . It is multiplicative in a and in n , it depends only on a modulo n , and it is non-zero exactly when a and n are coprime. We use the law of quadratic reciprocity for Jacobi symbols together with its supplements [1, Chapter 5, §2]: for odd positive integers m and n ,

$$\begin{aligned} \left(\frac{-1}{n}\right) &= (-1)^{(n-1)/2}, & \left(\frac{2}{n}\right) &= (-1)^{(n^2-1)/8}, \\ \left(\frac{m}{n}\right) \left(\frac{n}{m}\right) &= (-1)^{(m-1)(n-1)/4} \quad \text{if } \gcd(m, n) = 1. \end{aligned}$$

Lemma 2. *Let p be an odd prime with $(-d/p) = -1$, and let $a, b \in \mathbb{Z}$. If $p \mid a^2 + db^2$, then $p \mid a$ and $p \mid b$.*

Proof. Suppose that $p \nmid b$. Choose $c \in \mathbb{Z}$ with $bc \equiv 1 \pmod{p}$. Then $(ac)^2 \equiv -d(bc)^2 \equiv -d \pmod{p}$, so $-d$ is a square modulo p . As $(-d/p) \neq 0$, it is a non-zero square, so $(-d/p) = 1$, a contradiction. Hence $p \mid b$, and then $p \mid a^2$, so $p \mid a$. $\square$

Lemma 3. *Let $a, b \in \mathbb{Z}$ be not both zero. Then $a^2 + db^2 \neq 0$, and $v_p(a^2 + db^2)$ is even for every odd prime p with $(-d/p) = -1$.*

Proof. If $d > 0$, then clearly $a^2 + db^2 > 0$. If $d < 0$, then $-d > 1$ is squarefree, hence not a square, so $a^2 + db^2 \neq 0$. Let p be an odd prime with $(-d/p) = -1$, let p^k be the largest power of p dividing both a and b , and write $a = p^k a'$ and $b = p^k b'$. Then p does not divide both a' and b' , so $p \nmid a'^2 + db'^2$ by Lemma 2. Since $a^2 + db^2 = p^{2k}(a'^2 + db'^2)$, we get $v_p(a^2 + db^2) = 2k$. $\square$

Lemma 4. *Let n and n' be odd positive integers with $n' \equiv n \pmod{4|d|}$, or, if $d < 0$, with $n' \equiv -n \pmod{4|d|}$. Then $(-d/n') = (-d/n)$.*

Proof. Since d is squarefree, we can write $-d = \varepsilon 2^e q$ with $\varepsilon \in \{1, -1\}$, $e \in \{0, 1\}$ and q odd and positive. By multiplicativity and the reciprocity laws,

$$\left(\frac{-d}{n}\right) = \left(\frac{\varepsilon}{n}\right) \left(\frac{2}{n}\right)^e (-1)^{(n-1)(q-1)/4} \left(\frac{n}{q}\right).$$

Here (ε/n) and $(-1)^{(n-1)(q-1)/4}$ depend only on n modulo 4, $(2/n)$ depends only on n modulo 8, and (n/q) depends only on n modulo q . Since $4|d|$ is divisible by $4q$, and by $8q$ if $e = 1$, the claim for $n' \equiv n$ follows. If $d < 0$ and $n' \equiv -n \pmod{4|d|}$, then, as $\varepsilon = 1$, the factor (ε/n) does not change, $(2/n)$ does not change because $n'^2 \equiv n^2 \pmod{16}$, and each of $(-1)^{(n-1)(q-1)/4}$ and (n/q) is multiplied by $(-1)^{(q-1)/2}$, because $(n' - 1)/2 \equiv (n - 1)/2 + 1 \pmod{2}$ and $(n'/q) = (-1/q)(n/q) = (-1)^{(q-1)/2}(n/q)$. $\square$

Lemma 5. *Let n be a non-zero integer, and let s, u and m be positive integers such that every prime factor of s divides $2|d|$, both $2s$ and $4|d|s$ divide m , u is odd, $(-d/u) = -1$, and $n \equiv su \pmod{m}$. If $d > 0$, suppose also that $n > 0$. Then there is an odd prime p with $(-d/p) = -1$ such that $v_p(n)$ is odd.*

Proof. Since $s \mid m$, we have $s \mid n$, and $n' = n/s$ satisfies $n' \equiv u \pmod{m/s}$. Since m/s is even and divisible by $4|d|$, n' is odd and $n' \equiv u \pmod{4|d|}$. If $d > 0$, then $n' > 0$, and if $d < 0$, then $|n'| \equiv \pm u \pmod{4|d|}$; in both cases Lemma 4 gives $(-d/|n'|) = (-d/u) = -1$. By multiplicativity, the product of $(-d/p)^{v_p(n')}$ over the prime factors p of $|n'|$ is -1 , so some prime factor p of n' satisfies $(-d/p) = -1$ and $v_p(n')$ is odd. Then p is odd, and $p \nmid 2d$, so $p \nmid s$ and $v_p(n) = v_p(n')$ is odd. $\square$

2 The auxiliary polynomials

Put $F(x, y) = x^4 + 6x + y^3 + y - 1$, so that (1) is the equation $F(x, y) = 0$. For the first identity, $d_1 = 2$, and we define

$$\begin{aligned} H_1 &= -2x - y + 4, \\ T_1 &= 4x^2 - 2xy - 16x + y^2 + 4y + 17, \\ U_1 &= x^2 - 4x + 7, \\ V_1 &= 3x - 3, \\ G_1 &= -1, \\ A_1 &= 3, \\ B_1 &= -x + 3, \end{aligned}$$

for the second identity, $d_2 = 1$, and we define

$$\begin{aligned} H_2 &= -2x - y + 6, \\ T_2 &= 4x^2 - 2xy - 24x + y^2 + 6y + 37, \\ U_2 &= x^2 - 4x + 10, \\ V_2 &= 6x - 11, \\ G_2 &= -1, \\ A_2 &= 36, \\ B_2 &= -6x + 13, \end{aligned}$$

for the third identity, $d_3 = 2$, and we define

$$H_3 = -2y + 5,$$

$$\begin{aligned}
T_3 &= 8y^2 + 20y + 58, \\
U_3 &= 4x^2 - 16, \\
V_3 &= 8x + 3, \\
G_3 &= -16, \\
A_3 &= -16, \\
B_3 &= 8x - 3,
\end{aligned}$$

and, for the signs, we also use

$$\begin{aligned}
Q_1 &= 4x^2 - 2xy - 16x + y^2 + 4y + 17, \\
M_1 &= x^4 - 8x^3 + 48x^2 - 92x + 67, \\
E_1 &= 12x^2 - 48x + 52, \\
Q_2 &= 4x^2 - 2xy - 24x + y^2 + 6y + 37, \\
M_2 &= x^4 - 8x^3 + 72x^2 - 212x + 221, \\
E_2 &= 12x^2 - 72x + 112, \\
Q_3 &= 4y^2 + 10y + 29, \\
M_3 &= 8x^4 + 48x + 137, \\
E_3 &= 364.
\end{aligned}$$

Expanding both sides, one checks the identities

$$H_1T_1 - U_1^2 - 2V_1^2 = G_1F, \quad (2)$$

$$H_2T_2 - U_2^2 - V_2^2 = G_2F, \quad (3)$$

$$H_3T_3 - U_3^2 - 2V_3^2 = G_3F, \quad (4)$$

$$A_1U_1 + B_1V_1 = 2^2 \cdot 3, \quad (5)$$

$$A_2U_2 + B_2V_2 = 7 \cdot 31, \quad (6)$$

$$A_3U_3 + B_3V_3 = 13 \cdot 19, \quad (7)$$

$$-F = Q_1H_1 - M_1, \quad (8)$$

$$4Q_1 = (-2x + 2y + 4)^2 + E_1. \quad (9)$$

$$-F = Q_2H_2 - M_2, \quad (10)$$

$$4Q_2 = (-2x + 2y + 6)^2 + E_2. \quad (11)$$

$$-8F = Q_3H_3 - M_3, \quad (12)$$

$$16Q_3 = (8y + 10)^2 + E_3. \quad (13)$$

3 Proof of Theorem 1

Suppose that $(x, y) \in \mathbb{Z}^2$ is a solution of (1). Then $F(x, y) = 0$, and (2), (3), and (4) give

$$H_kT_k = U_k^2 + d_kV_k^2 \quad (k = 1, 2, 3), \quad (14)$$

where the polynomials now denote their values at (x, y) .

Step 1: congruences. Checking the 128^2 residue pairs modulo 128, we find that (1) has exactly 192 solutions modulo 128, and that at each of them $H_1 \bmod 128$ lies in the set $\mathcal{R}_1$ listed in Table 1, $H_2 \bmod 128$ lies in the set $\mathcal{R}_2$ listed in Table 1, or $H_3 \bmod 128$ lies in the set $\mathcal{R}_3$ listed in Table 1. For $k = 1$, each $r \in \mathcal{R}_1$ is listed as $r = su$ with s and u positive and u odd,

or as r if $s = 1$, and in each case $(-2/u) = -1$; here every prime factor of each $s \in \{1, 8, 16\}$ divides 4, and $2s$ and $8s$ divide 128 for each such s . For $k = 2$, each $r \in \mathcal{R}_2$ is listed as $r = su$ with s and u positive and u odd, or as r if $s = 1$, and in each case $(-1/u) = -1$; here for $s = 1$, $2s = 2$ and $4s = 4$ divide 128. For $k = 3$, each $r \in \mathcal{R}_3$ is listed as $r = su$ with s and u positive and u odd, or as r if $s = 1$, and in each case $(-2/u) = -1$; here for $s = 1$, $2s = 2$ and $8s = 8$ divide 128.

Step 2: signs. *The sign of H_1 .* Since $F(x, y) = 0$, (8) gives $Q_1H_1 = M_1$, and M_1 and E_1 depend only on x . We show that $M_1 > 0$ and $E_1 > 0$. If $x \geq 2$, write $x = 2 + z$ with $z \geq 0$; if $x \leq -1$, write $x = -1 - z$ with $z \geq 0$. In both cases, expanding M_1 and E_1 as polynomials in z gives non-negative coefficients and positive constant terms, so $M_1 > 0$ and $E_1 > 0$. It remains to consider $0 \leq x \leq 1$. For $x = 0$ and $x = 1$ we have $M_1 > 0$ and $E_1 > 0$. Hence $M_1 > 0$ and $E_1 > 0$, so (9) gives $4Q_1 \geq E_1 > 0$, and $Q_1H_1 = M_1 > 0$ gives $H_1 > 0$.

The sign of H_2 . Since $F(x, y) = 0$, (10) gives $Q_2H_2 = M_2$, and M_2 and E_2 depend only on x . We show that $M_2 > 0$ and $E_2 > 0$. If $x \geq 3$, write $x = 3 + z$ with $z \geq 0$; if $x \leq -1$, write $x = -1 - z$ with $z \geq 0$. In both cases, expanding M_2 and E_2 as polynomials in z gives non-negative coefficients and positive constant terms, so $M_2 > 0$ and $E_2 > 0$. It remains to consider $0 \leq x \leq 2$. For $x \in \{0, 1, 2\}$ we have $M_2 > 0$ and $E_2 > 0$. Hence $M_2 > 0$ and $E_2 > 0$, so (11) gives $4Q_2 \geq E_2 > 0$, and $Q_2H_2 = M_2 > 0$ gives $H_2 > 0$.

The sign of H_3 . Since $F(x, y) = 0$, (12) gives $Q_3H_3 = M_3$, and M_3 and E_3 depend only on x . We show that $M_3 > 0$ and $E_3 > 0$. If $x \geq 1$, write $x = 1 + z$ with $z \geq 0$; if $x \leq -2$, write $x = -2 - z$ with $z \geq 0$. In both cases, expanding M_3 and E_3 as polynomials in z gives non-negative coefficients and positive constant terms, so $M_3 > 0$ and $E_3 > 0$. It remains to consider $-1 \leq x \leq 0$. For $x = -1$ and $x = 0$ we have $M_3 > 0$ and $E_3 > 0$. Hence $M_3 > 0$ and $E_3 > 0$, so (13) gives $16Q_3 \geq E_3 > 0$, and $Q_3H_3 = M_3 > 0$ gives $H_3 > 0$.

Step 3: primes p with $(-d_k/p) = -1$. Let $k \in \{1, 2, 3\}$, and let p be an odd prime with $(-d_k/p) = -1$, where $(d_1, d_2, d_3) = (2, 1, 2)$. We show that $v_p(H_k)$ is even. By (5)–(7), U_k and V_k are not both zero, so (14) and Lemma 3 show that $v_p(H_kT_k) = v_p(U_k^2 + d_kV_k^2)$ is even. Suppose that p divides both H_k and T_k . Then $p \mid U_k^2 + d_kV_k^2$ by (14), so $p \mid U_k$ and $p \mid V_k$ by Lemma 2, and $p \mid R_k$, where R_k is the right-hand side of the identity for $A_kU_k + B_kV_k$. For $k = 1$, we have $R_1 = 2^2 \cdot 3$. But p is odd, and $p \neq 3$ because $(-2/3) = 1$, a contradiction. For $k = 2$, we have $R_2 = 7 \cdot 31$. As p is odd, we get $p \in \{7, 31\}$. Checking the 7^2 residue pairs modulo 7 and the 31^2 residue pairs modulo 31, we find that F , H_2 and T_2 have no common zero modulo 7 and 31, a contradiction. For $k = 3$, we have $R_3 = 13 \cdot 19$. As p is odd and $p \neq 19$ because $(-2/19) = 1$, we get $p = 13$. Checking the 13^2 residue pairs modulo 13, we find that F , H_3 and T_3 have no common zero modulo 13, a contradiction. Hence p does not divide both H_k and T_k . If $p \nmid T_k$, then $v_p(H_k) = v_p(H_kT_k)$ is even, and if $p \nmid H_k$, then $v_p(H_k) = 0$.

Step 4: contradiction. By Step 1, there is k such that $H_k \equiv su \pmod{128}$ with s and u as in Lemma 5 for $d = d_k$. By Step 2, $H_k \neq 0$, and $H_k > 0$ if $d_k > 0$. Hence Lemma 5 applies to $n = H_k$ and $d = d_k$, so there is an odd prime p with $(-d_k/p) = -1$ such that $v_p(H_k)$ is odd. This contradicts Step 3, and the theorem is proved. $\square$

References

- [1] K. Ireland and M. Rosen, *A Classical Introduction to Modern Number Theory*, 2nd ed., Graduate Texts in Mathematics 84, Springer, New York, 1990.

Table 1: The sets $\mathcal{R}_k$ of residues modulo 128 in Step 1.

$\mathcal{R}_1$, the residues of H_1					
5	29	53	71	93	$112 = 16 \cdot 7$
7	31	55	77	95	117
13	37	$56 = 8 \cdot 7$	79	101	119
15	39	61	$80 = 16 \cdot 5$	103	$120 = 8 \cdot 15$
21	45	63	85	109	125
23	47	69	87	111	127
$\mathcal{R}_2$, the residues of H_2					
3	27	51	75	99	123
11	35	59	83	107	
19	43	67	91	115	
$\mathcal{R}_3$, the residues of H_3					
15	47	79	111		
31	63	95	127		